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BATCH - T11

ROLL NO. - 13

EXPERIMENT 4

Aim : To understand Continuous Integration, install and configure jenkins with maven/ant/gradle to setup a build job

Theory :

Continuous Integration (CI):

Continuous Integration is a software development practice where code changes are automatically built, tested, and integrated into a shared repository on a frequent basis. The primary goal of CI is to detect and address integration issues early in the development process, ensuring that the codebase remains stable and reliable. This approach allows teams to deliver high-quality software more efficiently and with greater confidence.

Key Concepts of Continuous Integration:

1. Version Control System (VCS): CI relies on a VCS (e.g., Git, SVN) to manage and track changes in the codebase. Developers commit their changes to the repository, triggering the CI process.
2. Automated Build: CI systems automate the process of compiling source code into executable artifacts. This ensures consistency and eliminates manual errors in the build process.
3. *Automated Testing:* CI includes automated testing to validate that code changes do not introduce new bugs or break existing functionality. Common types of tests include unit tests, integration tests, and acceptance tests.
4. Build Server: A CI server, like Jenkins, is responsible for orchestrating the CI process. It monitors the VCS for changes, triggers builds, runs tests, and provides feedback to developers.

Installing and Configuring Jenkins with Maven/Ant/Gradle:

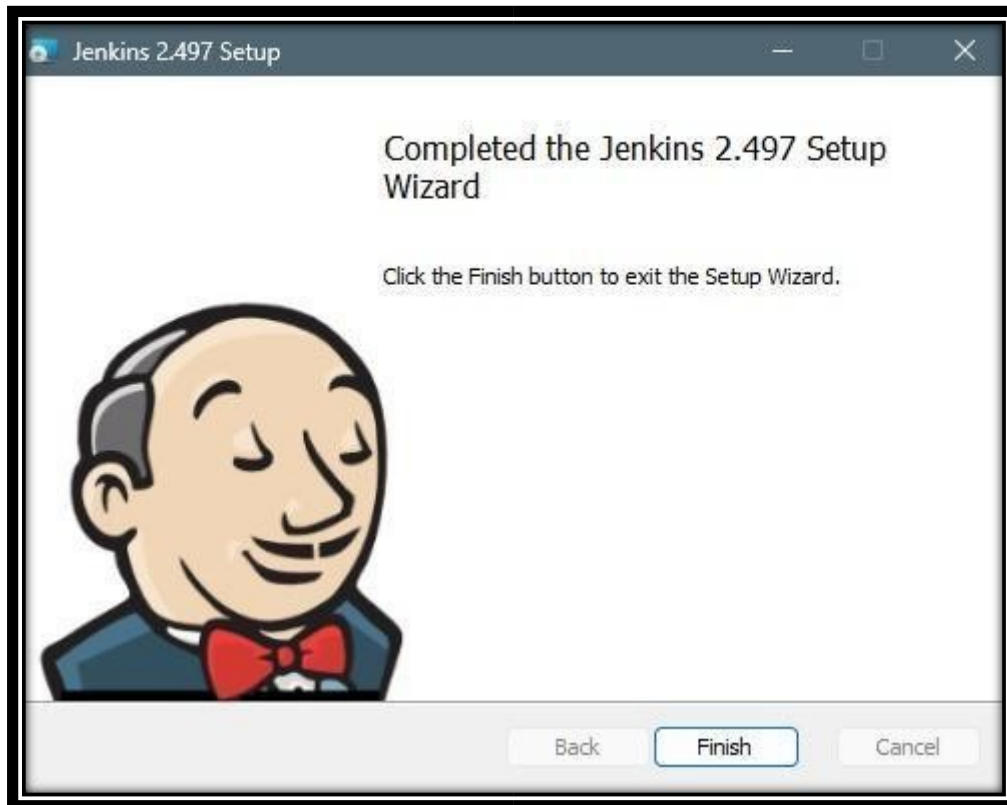
Here, we'll focus on setting up Jenkins with Maven, but the process is similar for other build tools.

1. Install Jenkins:

- Download and install Jenkins from the official website.



- Start the Jenkins service.



2. Configure Jenkins:

- Open Jenkins in a web browser and follow the initial setup wizard. - Install necessary plugins, including the ones for Maven, Ant, or

Gradle integration.

3. Create a Jenkins Job:

- Click on "New Item" to create a new job.

- Choose the type of project (e.g., Freestyle project or Pipeline). - Configure the job settings, such as source code repository, build triggers, and post-build actions.

4. *Configure Build Tool:*

- If using Maven, specify the path to the Maven executable and configure Maven goals (e.g., clean install).

- For Ant or Gradle, configure the respective build tool settings.

5. *Save and Build:*

- Save the job configuration and manually trigger the build to test the setup. - Observe the build console output for any errors or issues.

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Tomcat
12-Feb-2025 11:44:46.716 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -XX:+E
xplicitGCInvokesConcurrent
12-Feb-2025 11:44:46.716 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Xms38
4m
12-Feb-2025 11:44:46.716 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Xmx20
48m
12-Feb-2025 11:44:46.716 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Digno
re.endorsed.dirs=
12-Feb-2025 11:44:46.716 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Dcata
lina.base=C:\Users\15L\Atlassian\Jira
12-Feb-2025 11:44:46.717 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Dcata
lina.home=C:\Users\15L\Atlassian\Jira
12-Feb-2025 11:44:46.717 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Djava
.io.tmpdir=C:\Users\15L\Atlassian\Jira\temp
12-Feb-2025 11:44:46.775 INFO [main] org.apache.catalina.core.AprLifecycleListener.lifecycleEvent Loaded Apache Tomcat N
ative library [1.3.1] using APR version [1.7.4].
12-Feb-2025 11:44:46.775 INFO [main] org.apache.catalina.core.AprLifecycleListener.lifecycleEvent APR capabilities: IPv6
[true], sendfile [true], accept filters [false], random [true], UDS [true].
12-Feb-2025 11:44:46.775 INFO [main] org.apache.catalina.core.AprLifecycleListener.lifecycleEvent APR/OpenSSL configurat
ion: useAprConnector [false], useOpenSSL [true]
12-Feb-2025 11:44:46.786 INFO [main] org.apache.catalina.core.AprLifecycleListener.initializeSSL OpenSSL successfully in
itialized [OpenSSL 3.0.14 4 Jun 2024]
12-Feb-2025 11:44:47.005 INFO [main] org.apache.coyote.AbstractProtocol.init Initializing ProtocolHandler ["http-nio-808
0"]
12-Feb-2025 11:44:47.014 INFO [main] org.apache.catalina.startup.Catalina.load Server initialization in [521] millisecon
ds
12-Feb-2025 11:44:47.032 INFO [main] org.apache.catalina.core.StandardService.startInternal Starting service [Catalina]
12-Feb-2025 11:44:47.032 INFO [main] org.apache.catalina.core.StandardEngine.startInternal Starting Servlet engine: [Apa
che Tomcat/9.0.98]
```

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___ Starting the JIRA Plugin System _____

2025-02-12 11:46:16,567+0530 JIRA-Bootstrap INFO [c.a.j.config.database.SystemDatabaseConfigurationLoader] Reading
database configuration from C:\Users\15L\Atlassian\Application Data\Jira\dbconfig.xml
2025-02-12 11:46:16,594+0530 JIRA-Bootstrap INFO [c.a.jira.startup.JiraStartupLogger] Running Jira startup checks.
2025-02-12 11:46:16,595+0530 JIRA-Bootstrap INFO [c.a.jira.startup.JiraStartupLogger] Jira pre-database startup che
cks completed successfully.
2025-02-12 11:46:16,667+0530 JIRA-Bootstrap INFO [c.a.j.config.database.SystemDatabaseConfigurationLoader] Reading
database configuration from C:\Users\15L\Atlassian\Application Data\Jira\dbconfig.xml
2025-02-12 11:46:16,667+0530 JIRA-Bootstrap INFO [c.a.j.config.database.DatabaseConfigurationManagerImpl] The datab
ase is not yet configured. Enqueuing Database Checklist Launcher on post-database-configured-but-pre-database-activated
queue
2025-02-12 11:46:16,667+0530 JIRA-Bootstrap INFO [c.a.j.config.database.DatabaseConfigurationManagerImpl] The datab
ase is not yet configured. Enqueuing Post database-configuration launchers on post-database-activated queue
2025-02-12 11:46:16,680+0530 JIRA-Bootstrap INFO [c.a.jira.startup.LauncherContextListener] Startup is complete. Ji
ra is ready to serve.
2025-02-12 11:46:16,685+0530 JIRA-Bootstrap INFO [c.a.jira.startup.LauncherContextListener] Memory Usage:

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Heap memory      : Used: 269 MiB. Committed: 384 MiB. Max: 2048 MiB
Non-heap memory  : Used: 103 MiB. Committed: 143 MiB. Max: 1536 MiB
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TOTAL            : Used: 372 MiB. Committed: 527 MiB. Max: 3584 MiB
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```

CONCLUSION : Successfully understood Continuous Integration, installed and configured jenkins with maven/ant/gradle